Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [MSCCH204]

SECOND SEMESTER UNIVERSITY EXAMINATION, JUNE 2016

M.Sc. (CHEMISTRY)

SPECTROSCOPY-I

Time : 03 Hours **Maximum Marks:80**

Instructions: 1. Questions 1 to 6 carry 15 marks each, answer any four.

2. Question 7 is compulsory, carries 20 marks.

1. a. Deduce the vibrational mode representations for water molecule by shortcut (matrix) method, identify the irreducible components and explain the nature of vibration with the help of irreducible components.

Given: C_{2v} point group

C_{2v}	E	C_{2z}	σ_{xz}	σ_{yz}	Linear functions, rotations	Quadratic functions
A_1	+1	+1	+1	+1	z	x^2, y^2, z^2
A_2	+1	+1	-1	-1	R_z	xy
B_1	+1	-1	+1	-1	x, R_y	xz
B_2	+1	-1	-1	+1	y, R_x	yz

b. State "symmetry elements" and "symmetry operations" and explain with suitable examples. **10+5**

2. a. What are the differences between Raman and IR spectroscopy?

b. Write a note on reducible and irreducible representations in group theory. **10+5**

3. a Calculate the approximate wavenumber for C-O, C-H, C=C and O-H stretching in IR

Given: Force constant = 5×10^5 dynes/cm, mass of carbon = 12, mass of oxygen = 16, mass of hydrogen = 1 and $N_A = 6.02 \times 10^{23}$.

b. What are the three conditions for a molecule to be IR active? Whether symmetric stretching of CO_2 molecule is IR active? Explain. **10+5**

4. a. What is simplification of complex NMR spectrum? Discuss the various methods of simplifying complex NMR spectrum.

b. What is magnetic anisotropy in NMR spectroscopy? Explain with an example. **10+5**

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5. a. Write the mass fragmentation pattern of primary, secondary and tertiary alcohols.

b. Write the fragmentation and mass spectrum of $R\text{-}CH_2\text{-}NH_2$ and $R\text{-}CH_2\text{-}CH_2\text{-}OH$.

c. Explain the Matrix Assisted Laser Desorption Ionisation (MALDI) technique. **3×5**

6. a. An organic compound with molecular formula $C_9H_{10}O_2$ showed following spectral data

IR (cm^{-1}): 1740 (s), 1610-1400 (multiple bands), 1220, 750, 700

PMR δ: 1.96 (s, 3H); 5.0 (s, 2H); 7.2 (s, 5H)

Mass m/z: 43; 91; 150.

Deduce the structure and account for mass fragmentation pattern.

b. When acetone is treated with base, a higher boiling liquid (bp: 130°C) can be isolated from reaction mixture. The spectroscopic data of this liquid are

IR (cm^{-1}): 1620(m), 1695(s)

PMR: δ 1.9 (s, 3H); 2.1 (s, 6H); 6.15 (s, 1H)

UV: λ_{max}11700

Mass: m/z: 39, 43, 53, 55, 83, 98

^{13}CNMR: δ: 20, 27, 31, 124, 154, and 197.

Deduce the structure, account for all the observed data and write down the organic transformation. **5+10**

7. **Answer any TEN of the following.** **10×2=20**

a. Write down 3×3 matrix representation for C_{2z} axis.

b. Find the point groups for pyridine and boric acid molecules.

c. Write down the Mulliken Symbols for the following irreducible representations

C_{2h}	E	C_{2z}	i	σ_{xz}
Γ_1	1	1	-1	-1
Γ_2	1	-1	1	-1

d. Explain electronic transitions in UV spectroscopy.

e. Explain Hooks law.

f. Why homonuclear diatoms are microwave inactive?

g. Calculate the resonating energy for proton. Given: $\gamma = 2.675 \times 10^8$ rads^{-1}T^{-1}, $B_0 = 1.4$ T

h. Explain the effect of electronegative substitution in ^{13}C NMR spectroscopy.

i. How sensitivity is defined in mass spectrometry?

j. Sketch the basic components of a mass spectrometer.

k. An organic compound has molecular formula C_3H_6O, calculate the Index of Hydrogen Deficiency (IHD) and draw all probable structures of the molecule.

l. An organic compound with even molecular weight showed three signals in ^1H NMR at δ 3.5 (t, 2H), 1.5 (m, 2H) and 0.9 (t, 3H) in D_2O solvent and its IR spectrum showed broad and strong band at 3300 cm^{-1}. Predict the product.

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